MODULE 05 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Edge Architecture and Multi-Tenant Isolation

Code that runs before the request travels, and the boundaries that keep one customer out of another's data.

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Code at the edge is for decisions that must happen **before the request travels** to your origin.

CloudFront Functions or Lambda@Edge

	CLOUDFRONT FUNCTIONS	LAMBDA@EDGE
Events	Viewer request and response	Viewer and origin, both directions
Duration	Sub-millisecond	Seconds
Code size	10 KB, JavaScript	Up to 50 MB, Node.js or Python
Network and body	No network, no body	Network calls and request body
Scale	Millions of requests per second	10,000 per second per Region
State	CloudFront KeyValueStore	Any AWS service via the SDK

ON THE EXAM

If the function needs the network, a library, the request body or an origin event, it is Lambda@Edge.

Which runtime for which job?

CLOUDFRONT FUNCTIONS

Lightweight request shaping

- Redirects and URL rewrites
- Add or change headers
- Normalise the cache key
- Validate a JWT signature

LAMBDA@EDGE

Real processing

- Call an external service or database
- Resize images on a cache miss
- Inspect or change the request body
- Use third-party libraries

THE TIEBREAKER

Default to CloudFront Functions. Move to Lambda@Edge only when a named capability forces it.

Edge patterns worth naming

- **Cache key normalisation** raises the hit ratio, which lowers latency and origin load.
- **KeyValueStore** holds redirects, feature flags or A/B weights without redeploying.
- **Origin selection** per request, from a function, for routing tenants or versions.
- **Reject bad requests at the edge** so invalid tokens never reach the origin.
- **Lambda@Edge is deployed from us-east-1** and replicated outwards.

> ON THE EXAM
>
> The cheapest request is the one your origin never sees. Edge logic is often a cost answer as well as a latency one.

In a shared system, the tenant ID is a **security boundary**. Everything that touches data must carry it and enforce it.

Silo, bridge and pool

MODEL	SHAPE	ISOLATION	COST PER TENANT
Silo	Dedicated stack or account per tenant	Strongest	Highest
Bridge	Shared compute, dedicated data stores	Strong for data	Medium
Pool	Everything shared, tenant ID in every row	Enforced by policy	Lowest

ON THE EXAM

Most real platforms are tiered: silo for the few customers who pay for it, pool for everyone else.

Carrying tenant context through every layer

LAYER	MECHANISM
Identity	Cognito: tenant attribute in tokens, or a pool per tenant
API	API Gateway usage plans per tenant tier
Compute	Lambda tenant isolation mode, since November 2025
Data	DynamoDB LeadingKeys, PostgreSQL row-level security
Encryption	KMS key per tenant, or encryption context
Queues	SQS fair queues, since July 2025

ON THE EXAM

Tenant context starts in a signed token. Every later layer trusts the token, never a header the client wrote.

Cognito: shared pool or pool per tenant?

SHARED POOL, TENANT ATTRIBUTE

Surface-level tenancy

- One sign-in for every tenant
- custom:tenantID, read-only to clients
- Same MFA and branding for all
- App enforces the claim everywhere

POOL PER TENANT

Strong identity isolation

- MFA policy set per tenant
- Own branding and identity providers
- No shared user directory
- More pools to provision and manage

THE TIEBREAKER

Settings that live at pool level, such as MFA, cannot vary by attribute. Different MFA rules mean different pools.

Enforcing isolation on pooled data

- **DynamoDB:** dynamodb:LeadingKeys condition restricts items to the caller's tenant partition.
- **Session tags** put the tenant ID in the role session; the policy reads aws:PrincipalTag.
- **Aurora PostgreSQL:** IAM cannot scope rows. Use row-level security policies.
- **S3:** a prefix per tenant, with policy variables restricting each to its own.
- **Application filters alone** are the weakest control: one missed WHERE clause leaks.

> ON THE EXAM
>
> Isolation enforced by IAM or the database survives an application bug. A filter in code does not.

Tenant-specific encryption

KMS KEY PER TENANT

Strong, per-tenant control

- Separate key policy for each tenant
- Disable the key to cut off one tenant
- Per-tenant audit in CloudTrail
- More keys, more cost, more quota

SHARED KEY, ENCRYPTION CONTEXT

Pooled, still auditable

- tenant_id passed as encryption context
- Policy condition on kms:EncryptionContext
- Context appears in CloudTrail
- One key for many tenants

THE TIEBREAKER

A tenant who must be able to revoke access, or who supplies their own key, needs a key of their own.

Noisy neighbours and shared compute

- **SQS fair queues:** set a message group ID on a standard queue; no consumer changes.
- **Usage plans** throttle each tenant tier at the API.
- **Reserved concurrency** per tier stops one tier starving another.
- **Lambda tenant isolation mode:** pass a tenant ID on invoke; environments never cross tenants.
- **A FIFO queue is not a fairness tool.** It orders, and it limits throughput.

> ON THE EXAM
>
> Fair queues keep standard-queue throughput while stopping one tenant's backlog delaying the rest.

CloudFront SaaS Manager

- **Multi-tenant distribution:** a template of shared origins, behaviours and security.
- **Distribution tenants** inherit it, each with its own domain and certificate.
- **Per-tenant overrides:** WAF web ACL, geographic restrictions, parameters such as origin path.
- **Connection groups** provide the routing endpoint that tenants' CNAMEs point to.
- **Not supported:** continuous deployment, OAI, dedicated IP SSL.

> ON THE EXAM
>
> Thousands of customer domains on one platform: one multi-tenant distribution, not thousands of distributions.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Headers, redirects, millions per second	CloudFront Functions
Resize, call a service, read the body	Lambda@Edge
A few customers need guaranteed separation	Tiered silo and pool
Enforced by IAM, not application code	LeadingKeys with session tags
One tenant's jobs delay everyone	SQS fair queues
Different MFA rules per tenant	Cognito user pool per tenant

ON THE EXAM

Multi-tenant questions reward the answer that enforces isolation below the application.

> Meridian Freight's customer site on CloudFront serves several million requests per second at peak. The company must add security headers to every response and redirect legacy /track URLs to /shipments.
>
> The solution should add as little latency and cost as possible.

What should the solutions architect implement?

A. Lambda@Edge functions on the origin request and origin response events

B. CloudFront Functions on the viewer request and viewer response events

C. Listener rules on the origin's Application Load Balancer

D. A Lambda function behind API Gateway that rewrites each request

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A. Lambda@Edge on origin events

Origin events only run on cache misses, so cached responses miss the headers. Costlier too.

B. CloudFront Functions on viewer events

Sub-millisecond, scales to millions per second, runs on every request and response.

C. ALB listener rules

Only sees cache misses, and redirects would travel all the way to the Region.

D. API Gateway and Lambda

Adds a regional hop and per-request cost to every request.

THE SIGNAL

Simple header and redirect logic at very high volume is the textbook CloudFront Functions case.

> Meridian Freight's mobile app requests proof-of-delivery photos in several sizes. Originals are stored once in Amazon S3. Resized versions should be created on first request and cached at the edge.
>
> The resizing uses a third-party image library and must fetch the original from S3.

How should the resizing be implemented?

A. A CloudFront Function on the viewer request event

B. S3 Object Lambda access points

C. A nightly job that pre-generates every size for every photo

D. A Lambda@Edge function on the origin response event

◆ PAUSE AND DECIDE

How should the resizing be implemented?

A CloudFront Function

No libraries, no network access, 10 KB of code. It cannot resize an image.

B S3 Object Lambda

Closed to new customers, and not an edge cache pattern.

C Nightly pre-generation

Creates sizes nobody requests, and new photos wait until the next night.

D Lambda@Edge on origin response

Runs only on a cache miss, can use the library and fetch from S3; the result is cached.

THE SIGNAL

A library and a network fetch rule out CloudFront Functions. Cache-miss-only work belongs on an origin event.

> Meridian Freight sells its tracking platform as SaaS. Three bank customers have contracts requiring that their data and compute never share infrastructure with other customers.
>
> About 400 smaller customers are price-sensitive, and the company must keep their cost to serve low.

Which tenancy model should the solutions architect recommend?

A. Pool every tenant, with row-level security in a shared database

B. A dedicated AWS account and stack for every tenant

C. Silo stacks in dedicated accounts for the three banks, and a pooled tier for everyone else

D. Pool every tenant, with a separate KMS key for each

◆ PAUSE AND DECIDE

Which tenancy model should the solutions architect recommend?

A **Pool everyone with RLS**

Good isolation, but the banks' compute and database are still shared. Breaks the contract.

B **Silo everyone**

Meets the banks' terms, but 400 dedicated stacks wreck the cost to serve.

C **Tiered silo and pool**

Physical separation where contracts demand it, shared economics everywhere else.

D **Pool with a key per tenant**

Separates keys, not infrastructure. The banks still share compute.

THE SIGNAL

Two populations with different requirements, in one scenario, is the signature of a tiered model.

> Meridian Freight's pooled tier stores all tenants' shipments in one DynamoDB table with tenantId as the partition key. Lambda functions currently filter by tenant in application code.
>
> After a near miss in code review, the security team wants tenant isolation enforced outside the application code, without a policy or table per tenant.

What should the solutions architect implement?

A. Assume a role with a TenantID session tag, using a policy with dynamodb:LeadingKeys set to that tag

B. A separate DynamoDB table for each tenant

C. Mandatory code review for every query that touches the table

D. A separate customer managed KMS key for each tenant

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Session tag and LeadingKeys

IAM rejects any request outside the tenant's partition, with one shared policy.

B Table per tenant

Strong, but moves the pool tier to a bridge model, which the team ruled out.

C More code review

Still relies on application code. The near miss shows this is not enough.

D Key per tenant

Encrypts data at rest; does not stop a valid session reading another tenant's items.

THE SIGNAL

Enforced outside the code, with no per-tenant policy: IAM conditions driven by a tag on the session.

> Meridian Freight's pooled tier uses one SQS standard queue for report-generation jobs from all tenants. When one large tenant submits a bulk import, other tenants' jobs wait for hours.
>
> The company wants fair processing across tenants without changing the consumer code or creating a queue per tenant.

What should the solutions architect do?

A Convert to a FIFO queue using the tenant ID as the message group ID

B Create a separate queue for each tenant and poll them in rotation

C Double the consumer fleet during bulk imports

D Send each message with the tenant ID as its message group ID on the existing standard queue

◆ PAUSE AND DECIDE

What should the solutions architect do?

A FIFO with tenant groups

Adds strict ordering and throughput limits nobody asked for, and is a new queue.

B Queue per tenant

Explicitly ruled out, and needs new consumer polling logic.

C More consumers

Shortens the wait but does not make it fair; cost rises with every import.

D Fair queue via message group ID

SQS serves quieter tenants first. Same queue, consumers unchanged.

THE SIGNAL

A message group ID on a standard queue turns on fairness. On a FIFO queue it turns on ordering.

> Meridian Freight's enterprise customers each require sign-in through their own SAML identity provider, and two of them require different MFA rules from everyone else.
>
> Smaller customers sign in with a username and password, and the company wants their onboarding kept simple.

Which Amazon Cognito design meets these requirements?

A One user pool for all customers, with a custom:tenantID attribute

B A user pool per enterprise customer, and one shared pool with a tenant attribute for smaller customers

C One identity pool per customer

D One user pool with a group per customer

◆ PAUSE AND DECIDE

Which Amazon Cognito design meets these requirements?

A One pool, tenant attribute

MFA is set for the whole pool, so two customers cannot have different rules.

B Pools per enterprise, shared for small

Pool-level settings vary where needed; small customers stay simple.

C Identity pool per customer

Identity pools issue AWS credentials. They do not authenticate users or set MFA.

D Groups per customer

Groups carry roles and claims, but MFA is still pool-wide.

THE SIGNAL

MFA rules that differ between tenants cannot live in one user pool. That single detail decides it.

What to carry out of this module

- **CloudFront Functions by default.** Lambda@Edge for network, libraries, body or origin events.
- **Silo, bridge, pool,** and usually a tier of each. Read for two customer populations.
- **Enforce below the code:** LeadingKeys, row-level security, per-tenant keys.
- **New since 2025:** SQS fair queues, Lambda tenant isolation mode, CloudFront SaaS Manager.
- **Pool-level settings such as MFA** force a Cognito pool per tenant.

> ON THE EXAM
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> Skill 1.1.13 is new to this exam. Older courses do not teach tenant isolation at all.